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Chapter 9

Trajectory Generation

During robot motion, the robot controller is provided with a steady stream of goal positions and velocities to track. This specification of the robot position as a function of time is called a **trajectory**. In some cases, the trajectory is completely specified by the task – for example, the end-effector may be required to track a known moving object. In other cases, as when the task is simply to move from one position to another in a given time, we have freedom to design the trajectory to meet these constraints. This is the domain of **trajectory planning**. The trajectory should be a sufficiently smooth function of time, and it should respect any given limits on joint velocities, accelerations, or torques.

In this chapter we consider a trajectory as the combination of a **path**, a purely geometric description of the sequence of configurations achieved by the robot, and a **time scaling**, 时间缩放 which specifies the times when those configurations are reached. We consider three cases: point-to-point straight-line trajectories in both joint space and task space; trajectories passing through a sequence of timed **via points**; and minimum-time trajectories along specified paths taking actuator limits into consideration. Finding paths that avoid obstacles is left to Chapter 10.

path + 时间缩放 → trajectory → controller

1. 点→点直线 (joint space , task space)
2. 经过带时间戳的时间点轨迹
3. 考虑actuator限制，指定路径最小时间轨迹

9.1 Definitions

路径 构型 曲几何 0-1

A **path** $\theta(s)$ maps a scalar path parameter s, assumed to be 0 at the start of the path and 1 at the end, to a point in the robot's configuration space Θ, $\theta : [0, 1] \to \Theta$. As s increases from 0 to 1, the robot moves along the path. Sometimes s is taken to be time and is allowed to vary from time $s = 0$ to the total motion time $s = T$, but it is often useful to separate the role of the

轨迹 解耦 → 路径 $\theta(s)$ 时间缩放 $s(t)$ 可分别优化

时间缩放

geometric path parameter s from the time parameter t. A **time scaling** $s(t)$ assigns a value s to each time $t \in [0, T]$, $s : [0, T] \to [0, 1]$.

Together, a path and a time scaling define a **trajectory** $\theta(s(t))$, or $\theta(t)$ for short. Using the chain rule, the velocity and acceleration along the trajectory can be written as

$$\dot{\theta} = \frac{d\theta}{ds}\dot{s}, \tag{9.1}$$

$$\ddot{\theta} = \frac{d\theta}{ds}\ddot{s} + \frac{d^2\theta}{ds^2}\dot{s}^2. \tag{9.2}$$

To ensure that the robot's acceleration (and therefore dynamics) is well defined, each of $\theta(s)$ and $s(t)$ must be twice differentiable.

$\theta(s)$, $s(t)$ 都应二阶可微

9.2 Point-to-Point Trajectories

The simplest type of motion is from rest at one configuration to rest at another. We call this a point-to-point motion. The simplest type of path for point-to-point motion is a straight line. Straight-line paths and their time scalings are discussed below.

9.2.1 Straight-Line Paths

A "straight line" from a start configuration θ_{start} to an end configuration θ_{end} could be defined in joint space or in task space. The advantage of a straight-line path from θ_{start} to θ_{end} in joint space is simplicity: since joint limits typically take the form $\theta_{i,\min} \le \theta_i \le \theta_{i,\max}$ for each joint i, the allowable joint configurations form a convex set Θ_{free} in joint space, so the straight line between any two endpoints in Θ_{free} also lies in Θ_{free}. The straight line can be written

凸集性质 → 凸集

一阶 $\frac{d\theta}{ds} = \theta_{end} - \theta_{start}$

二阶 $\frac{d^2\theta}{ds^2} = 0$ (直线)

$$\theta(s) = \theta_{\text{start}} + s(\theta_{\text{end}} - \theta_{\text{start}}), \qquad s \in [0, 1] \tag{9.3}$$

$\ddot{\theta} = (\theta_{end} - \theta_{start})\ddot{s}$

with derivatives

$$\frac{d\theta}{ds} = \theta_{\text{end}} - \theta_{\text{start}}, \tag{9.4}$$

$$\frac{d^2\theta}{ds^2} = 0. \tag{9.5}$$

$\dot{\theta} = (\theta_{end} - \theta_{start})\dot{s}$

Straight lines in joint space generally do not yield straight-line motion of the end-effector in task space. If task-space straight-line motions are desired, the start and end configurations can be specified by X_{start} and X_{end} in task space. If